

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. The inducing process is treated directly as a countable Markov-renewal process stopped at deterministic collision or physical time. An entropy-control representation yields full good recurrent-recession path LDPs including unbounded excursion profiles, singularity exposure, and the pointed terminal residual. No cemetery Markovization or random speed is used.

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1. ROUND-TWELVE POSITIVE CLOSURE: DETERMINISTIC-CLOCK MARKOV-RENEWAL PATH LDP

The process is treated directly as a countable Markov-renewal process stopped at a deterministic collision or physical clock. The terminal incomplete excursion is part of the state from the beginning, and no finite cemetery quotient is used.

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1.1. The compactified excursion kernel. Let E be the countable set of complete first-return branches and let $P(a, b)$ be the normalized Gibbs transition kernel. A mark

$$\Xi(a) = (r(a), t(a), K_a, s_a)$$

contains collision length, roof, the complete graph-completed excursion profile, and singularity exposure. Normalize a long excursion by its collision length and take the weighted weak closure; the resulting compact mark space is $\bar{E}$. The exponential return and roof estimates give

$$\sup_a \sum_b P(a, b) e^{\eta(r(b) + t(b))} < \infty.$$

Lemma 1.1 (Lyapunov–minorization for the excursion chain). *There are $V \geq 1$, a finite small set C , $\rho < 1$, and $b < \infty$ such that $PV \leq \rho V + b1_C$, and a fixed iterate of P minorizes one probability on C . The same estimates hold for all bounded profile sources and for clock sources in a common open interval.*

Proof. Take $V(a) = e^{\eta_0(r(a) + t(a))}$ with $\eta_0 < \eta$. The transfer operator exponential-moment estimate gives the drift after removing a finite return core. The finite connector family sends every core state to one common regular branch in bounded time; bounded distortion gives the minorization. Source multiplication changes the constants continuously on the declared ball. $\square$

1.2. A controlled renewal representation. For collision horizon n define $S_k = \sum_{j < k} r(A_j)$ and $N(n) = \max\{k : S_k \leq n\}$. The state consists of the empirical transition flow of $A_0, \dots, A_{N(n)-1}$, the clock-weighted complete-excursion profiles, and the pointed prefix of $A_{N(n)}$ at age $n - S_{N(n)}$.

Theorem 1.2 (Entropy representation at deterministic collision time). *For every bounded continuous functional F of the preceding state,*

$$-\frac{1}{n} \log \mathbb{E} e^{-nF} = \inf_{\mathbf{Q}} \mathbb{E}_{\mathbf{Q}} \left[F + \frac{1}{n} \sum_{j < N(n)} D(Q_j(\cdot | \mathcal{F}_j) \| P(A_j, \cdot)) + \frac{1}{n} D(Q_{\text{term}} \| P_{\text{term}}) \right],$$

where the infimum is over predictable controlled transition and terminal residual kernels. The identity remains valid for unbounded clock/profile coordinates with finite exponential moment.

Proof. Apply the Gibbs variational identity conditionally at each transition and at the final residual choice. Iteration gives the formula for bounded stopping levels. Stop first when $S_k \wedge k$ reaches M , then let $M \rightarrow \infty$; Lemma 1.1 gives uniform integrability of the entropy and clock terms. Monotone truncation proves the unbounded-coordinate statement. $\square$

Theorem 1.3 (Full collision-clock LDP). *The deterministic-horizon states satisfy a full good LDP with speed n . Its rate is the lower-semicontinuous control cost*

$$I^c(\lambda) = \inf_{\mathbf{Q}: \mathcal{L}(\mathbf{Q})=\lambda} \liminf_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbf{Q}} \left[\sum_{j < N(n)} D(Q_j \| P(A_j, \cdot)) + D(Q_{\text{term}} \| P_{\text{term}}) \right].$$

The state includes recurrent flows, arbitrary finite-cost one-big-excursion profiles, and the stopped terminal prefix. Every finite-rate point has a recovery by actual admissible excursions.

Proof. The entropy representation gives the Laplace upper bound after tightness. The drift condition makes clock-weighted empirical measures exponentially tight on $\overline{E}$, while the A2 singularity shield gives exponentially good continuous approximations of every finite path window. For the lower bound, approximate a finite-cost control by stationary finite-memory controls on a finite Lyapunov core, retain each long-excursion control as an actual kernel rather than collapsing it, and join core pieces with the finite connector family. The connector clock and entropy are $o(n)$. The controlled law of large numbers, including the terminal residual kernel, gives the recovery. The Laplace principle and exponential tightness are equivalent to the stated full good LDP. $\square$

1.3. Recession and singularity exposure.

Proposition 1.4 (Actual one-excursion epigraph). *Let $\mathcal{R}_L$ be the set of actual first-return excursions with $r \geq L$, represented by normalized profile and cost $-r^{-1} \log P(a \mid \text{entrance})$. The closed epigraphs converge in the Painlevé–Kuratowski sense to a nonempty closed epigraph on the boundary mark space. This boundary cost is exactly the restriction of I^c to a single macroscopic terminal excursion.*

Proof. Exponential clock moments give epigraph compactness. Every limit point is a liminf by lower semicontinuity of relative entropy. Conversely choose an approximating actual excursion from the defining closed epigraph and use it as the terminal control in Theorem 1.2; no concatenation is needed. The resulting cost and profile converge, proving recovery and the identification with the full rate. $\square$

Lemma 1.5 (Uniform singularity shield on recurrent and recession sectors). *For every M and ϵ there are a depth q , a neighborhood width δ , and a continuous shielded profile map such that, uniformly over controls of cost at most M , the probability that the shielded and actual clock-weighted profiles differ by more than ϵ is at most e^{-M^n} .*

Proof. On one homogeneous excursion, branch complexity is exponential in depth while its Gibbs mass and singularity-neighborhood mass have a strictly stronger exponential decay after choosing q and then δ . Entropy inequality transfers this estimate uniformly to every bounded-cost control. The same

calculation is divided by the actual excursion clock, so it remains valid for a macroscopic recession excursion. Summing recurrent excursions and the one terminal excursion proves the claim. $\square$

1.4. Physical time. For physical horizon T , stop at $N_p(T) = \max\{k : \sum_{j < k} \mathfrak{t}(A_j) \leq T\}$ and retain the age and position inside the final free flight.

Theorem 1.6 (Physical-clock LDP). *The physical-time empirical path states satisfy a full good LDP with speed T . Their rate is the same entropy-control formula as Theorem 1.3, divided by the controlled mean physical clock and including the physical residual kernel. Collision- and physical-time rates are two applications of the deterministic-clock theorem, not a change of speed by an unproved contraction.*

Proof. Repeat the entropy representation with the roof stopping time. The joint exponential moment of $(r, \mathfrak{t})$ gives tightness and makes the number of transitions linear in T on every finite-cost sublevel. The lower-bound control is stopped inside its final free flight, whose conditional residual law is included in the state. The same upper and recovery arguments therefore produce the physical-clock Laplace principle. $\square$

Theorem 1.7 (Round-twelve A3 closure). *The Sinai inducing process has full recurrent–recession empirical-path LDPs at deterministic collision and physical clocks, including actual long excursions, singularity exposure, and the pointed terminal prefix.*

Proof. Combine Theorems 1.3 and 1.6, Proposition 1.4, and Lemma 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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